Jay Patrikar

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EDUCATION

Carnegie Mellon University | School Of Computer Science (Robotics Institute) *Pittsburgh, PA*
Doctor of Philosophy in Robotics *Fall 2020 – Summer 2026*[1]
Advisor: Dr. Sebastian Scherer

Carnegie Mellon University | School Of Computer Science (Robotics Institute) *Pittsburgh, PA*
Master of Science in Robotics *Fall 2018 – Summer 2020*
Advisor: Dr. Sebastian Scherer
GPA: 4.05 / 4.00

Indian Institute of Technology Kanpur | Department Of Aerospace Engineering *Kanpur, India*
Masters and Bachelors of Technology (Intergrated) *Fall 2013 – Summer 2018*
Advisor: Dr. Mangal Kothari
CPI (PG) : 10.0/10.0, CPI (UG): 8.6/10.0
Minors : Controls Systems (EE) and English Literature (Humanities)

Selected Courses: Planning and Decision-making in Robotics, Deep Reinforcement Learning, Computer Vision, Introduction to Machine Learning (PhD), Flight Mechanics, Integrated Intelligence in Robotics

RESEARCH EXPERIENCE

Airlab, *Carnegie Mellon University* *Pittsburgh, PA*
Director: Dr. Sebastian Scherer *2018 – Now*

- Worked on various projects that use artificial intelligence to improve the robustness and reliability of aerial autonomy.

Intelligent Guidance & Control Laboratory, *Indian Institute of Technology Kanpur* *Kanpur, India*
Director: Dr. Mangal Kothari *2016 – 2018*

- Developed and flight tested a novel guidance law for 3D path following of fixed-wing Unmanned Aerial Vehicles using input-based nested saturation techniques to improve the stability characteristics.

Unmanned Aerial Vehicles Laboratory, *Indian Institute of Technology Kanpur* *Kanpur, India*
Director: Dr. A.K.Ghosh *2015 – 2017*

- Designed and tested autopilot systems for fixed wing and rotary UAVs with applications in surveillance and target-tracking.

National Aerothermochemistry and Hypersonics Laboratory, *Texas A&M University* *College Station, TX*
Director: Dr. R Bowersox *Summer 2016*

- Studied Boundary Layer Transition on Elliptic cone geometries in Hypersonic flow using thermal flow visualization.

AWARDS & RECOGNITION

- **2 A+ grades** for exceptional performance in courses at Carnegie Mellon University
- **2 Academic Excellence Awards** (for distinctive academic performance in years 2016-17 and 2017-18) at Indian Institute of Technology Kanpur
- Awarded the **Summer Undergraduate Research Grant for Excellence (SURGE)** for Summer 2015
- **4 A+ grades** for exceptional performance in courses at Indian Institute of Technology Kanpur
- **All India Rank 2219** (amongst 13,00,000 students) in IIT-Joint Entrance Examination 2013

SKILLS & LICENSES

- **Languages :** English, Hindi, Marathi (Native), French (A1 Level Cerification)
- **Coding :** C++, Python, SQL
- **Tools and Libs :** OMPL, ROS, PX4, PostGres, PostGIS, Redis, TensorFlow
- **Licenses :** Remote Pilot Certificate FAA Part 107, Student Pilot Certificate Part 61

TEACHING EXPERIENCE

- **TA, 16-833: Robot Localization and Mapping ,** Robotics Institute, Carnegie Mellon University *Spring 2021*
- **TA, 16-384: Robot Kinematics and Dynamics ,** Robotics Institute, Carnegie Mellon University *Fall 2020*
- **TA, AE361A: Aeromodel Design & Fabrication ,** Department of Aerospace Engineering, IIT Kanpur *Spring 2018*

[1] Planned

SELECTED PROJECTS

Close-Proximity Safe and Seamless Operation of Manned and Unmanned Aircraft in Shared Airspace *Pittsburgh, PA*
U.S. Army AI Task Force PI: Dr. Sebastian Scherer, Co-PI: Dr. Jean Oh *2021*

- Developed end-to-end multi-modal (vision, language and context) system to achieve safe manned-unmanned vehicle teaming to improve the system performance and have each (robot/human) teammate learn from each other in various aircraft operations.
- Areas: Multi-modal AI, Trajectory Prediction, Decision Making and Planning

Improving Last-mile Autonomous Delivery in Urban Areas *Pittsburgh, PA*
U.S. Department of Energy PI: Dr. Constantine Samaras *2019*

- Carried out a flight data collection campaign to collect energy consumption data for a done-based system to train a first-principle physics-based energy model.
- Areas: First principles modeling, Autonomous path planning, Energy Sciences

Development of a Preflight Path Planning Tool for General Aviation Pilots *Pittsburgh, PA*
Near Earth Autonomy Inc., PI: Dr. Sebastian Scherer *2019*

- Developed a multi-objective risk-aware graph-based pre-flight planner for VFR (Visual-Flight Rules) cross-country flights
- Areas: Cloud Computing, GIS, Database Management, Graph-based planning, risk modelling

Development of UAV Dynamics Model and Study of Control Effectiveness *Hyderabad, India*
DRDO Centre for High Energy Systems and Sciences (CHESS) PI: Dr. Mangal Kothari *2018*

- Designed a full system dynamics simulator for fixed-wing UAVs and performed ablation studies to characterise control surface effectiveness.
- Areas: 6-DOF Simulations, Control Ablation Studies

Development of a High Altitude Surveillance Platform *Banglore, India*
TATA Advanced Systems Pvt Ltd PI: Dr. A.K. Ghosh *2016*

- Carried out system identification of a high-altitude fixed wing platform for surveillance operations at $> 4500m$ MSL.
- Areas: High Altitude Flight Mechanics, Design Optimization, Parameter Estimation, UAV Operations Management

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LEADERSHIP AND MENTORING

- **President** The Flying Club, Carnegie Mellon University *2021 – 2022*
- **Student Representative** Robotics Institute, Graduate Student Assembly, Carnegie Mellon University *2020 – 2021*
- **President** Society of Aerospace Engineers, IIT Kanpur *2016 – 2017*
- **Student Representative** Departmental Undergraduate Committee, IIT Kanpur *2015 – 2016*

PUBLICATIONS

Thesis

[1] **Wind-Field Estimation and Curvature Continuous Path Planning for Low Altitude Urban Aerial Mobility**
Master's Thesis, Tech. Report, CMU-RI-TR-20-30 *2020*

[2] **A 3D Guidance Law for Path Following**
Master's Thesis, Tech. Report *2018*

Papers

[3] **CVaR-based Flight Energy Risk Assessment for Multirotor UAVs using a Deep Energy Model**
J Patrikar*, *B Moon**, *A Choudhry**, *C Samaras, S Scherer* *2021*
International Conference on Robotics and Automation (ICRA)

[4] **Adaptive Tube Library for Safe Online Planning Under Unknown Tracking Performance** [2]
*C Ho, **J Patrikar**, R Bonatti, S Scherer* *2021*
Robotics: Science and Systems

[5] **In-flight positional and energy use dataset of package delivery quadcopter UAVs**
*T Rodrigues, **J Patrikar**, A Choudhry, J Feldgoise, V Arcot, A Gahlaut, S Lau, B Moon, B Wagner, S Matthews, S Scherer, C Samaras* *2020*
Nature Scientific Data

[6] **Wind and the City: Utilizing UAV-Based In-Situ Measurements for Estimating Urban Wind Fields**
***J Patrikar**, B Moon, S Scherer* *2020*
International Conference on Intelligent Robots and Systems (IROS)

[2] Submitted

[7] **Real-time Motion Planning of Curvature Continuous Trajectories for Urban UAV Operations in Wind**
J Patrikar, *V Dugar, V Arcot, S Scherer* *2020*
International Conference on Unmanned Aircraft Systems (ICUAS)

[8] **Nested Saturation Based Guidance Law for Unmanned Aerial Vehicles**
J Patrikar, *VR Makkapati, A Pattanaik, H Parwana, M Kothari* *2019*
Journal of Dynamic Systems, Measurement, and Control, ASME

[9] **Sequential Auto-Landing of Multiple UAVs using Control Constrained Path Following**
J Patrikar, *VR Makkapati, M Kothari* *2019*
AIAA Guidance, Navigation, and Control Conference

[10] **Convolutional Neural Network Based Sensors for Mobile Robot Relocalization**
J Patrikar*, *H Sinha**, *EG Dhekane**, *G Pandey, M Kothari* *2018*
23rd International Conference on Methods & Models in Automation & Robotics

[11] **A novel fully quaternion based nonlinear attitude and position controller**
H Parwana, ***J Patrikar***, *M Kothari* *2018*
AIAA Guidance, Navigation, and Control Conference

[12] **A Low-Cost Tilt-Augmented Quadrotor Helicopter : Modeling and Control**
M Bhargavapuri, ***J Patrikar***, *SR Sahoo, M Kothari* *2018*
International Conference on Unmanned Aircraft Systems (ICUAS)

[13] **MARAAL: A Low Altitude Long Endurance Solar Powered UAV For Surveillance and Mapping Applications**
VS Dwivedi, ***J Patrikar***, *A Addamane, AK Ghosh* *2018*
23rd International Conference on Methods & Models in Automation & Robotics

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PATENTS

[1] **Using CVaR-based metrics for energy risk assessments of UAV flights** [3]
Jay Patrikar, *Brady Moon, Arnav Choudhry, Sebastian Scherer and Constantinos Samaras* *2021*

[2] **Multi-camera visual detect & avoid system and detection algorithms.**
Sebastian Scherer, Sourish Ghosh, ***Jay Patrikar*** *and Brady Moon* *2021*
CMU Intellectual Property Disclosure no. 2021-113

[3] **Wind-Aware Planning for Last Mile Delivery**
Sebastian Scherer, ***Jay Patrikar****, Vishal Dugar, Vaibhav Arcot and Constantinos Samaras* *2020*
CMU Intellectual Property Disclosure no. 2020-071

[3] Submitted